A Machine Learning-Based Framework for Digital Addiction Risk Prediction with Integrated Health and Academic Performance Analysis among Teenagers

Yogita Y. V., Vaishnav Venkatesh, Dr. Sridevi Ponmalar
Department of Computational Intelligence, School of Computing
SRM Institute of Science and Technology, Chennai, India

Abstract—Widespread smartphone use has reshaped how adolescents spend their time and approach academic work. Although these devices bring clear benefits in terms of connectivity and access to learning material, prolonged and unregulated use has been linked to a cluster of behavioural concerns that are now broadly described as digital addiction. In this work we examine this phenomenon using a data-driven pipeline built around a short behavioural questionnaire. A total of 125 teenage respondents completed a 20-item instrument covering daily screen time, social-media exposure, night-time phone use, checking frequency, sleep duration, difficulty concentrating on studies, and self-reported academic standing. After preprocessing, four supervised classifiers—Random Forest, Logistic Regression, Decision Tree, and Support Vector Machine—were trained to place each respondent into one of three risk bands (low, moderate, high). Random Forest produced the strongest result, reaching 75% accuracy, and the feature-importance profile it learned placed concentration difficulty and average daily screen time at the top of the list. The framework is extended with a weighted health-risk module that estimates the likelihood of four adverse outcomes—eye strain, depressive symptoms, sleep disturbance, and anxiety—directly from the same behavioural features. Linking a predictive model with this health overlay yields a more rounded view of how heavy digital use affects teenagers, and makes the tool practical for educators and counsellors who need an early-warning indicator. Future extensions will widen the sample, introduce passively logged device-usage traces, and benchmark deep sequence models against the tree-based baselines reported here.

Index Terms—digital addiction, machine learning, random forest, adolescent behaviour, academic performance, health-risk assessment, smartphone overuse.

I. INTRODUCTION

Smartphones are no longer an accessory in a teenager's life; they have become the environment in which a large part of that life unfolds. Classwork, friendships, entertainment, and even sleep routines now revolve around a screen that is within arm's reach for most of the day. Against this backdrop, a subset of heavy users begins to show patterns that look less like ordinary habit and more like compulsive behaviour, and this cluster of symptoms is what the literature increasingly labels digital addiction.

The problem has drawn attention because the population at risk is still forming academic habits and coping strategies. Studies repeatedly report links between prolonged device use and three outcomes that matter for adolescents: lower grades, shorter and poorer-quality sleep, and elevated scores on scales measuring stress or low mood. What complicates the picture is that these effects feed back on one another—poor sleep degrades concentration, which pushes the student toward more passive screen time, which in turn delays sleep further.

Screening tools for this problem have historically relied on self-report questionnaires scored by fixed rules. Those instruments remain useful, but they treat the underlying indicators as independent and ignore the interactions between them. Machine-learning classifiers, by contrast, can pick up non-linear combinations of behavioural cues and rank them by predictive value, giving a more realistic weighting than a fixed Likert sum can. This is the gap the present study tries to fill.

One further dimension is often missing from purely behavioural work. Heavy smartphone use carries measurable physical and psychological costs—eye strain, disturbed sleep architecture, and heightened anxiety are the three most commonly cited in clinical reports. A screening tool that flags addiction risk without saying anything about these downstream health outcomes gives only half the picture. We therefore combine the classifier with a transparent health-risk layer so that any prediction is accompanied by an interpretable breakdown of likely consequences.

Concretely, this paper makes three contributions. First, we present a curated questionnaire covering behavioural, sleep, and academic items, administered to 125 Indian teenagers and released in anonymised form with the paper. Second, we benchmark four widely used classifiers—Random Forest, Logistic Regression, Decision Tree, and Support Vector Machine—on the three-class addiction-risk task and analyse the feature-importance profile of the winning model. Third, we add a weighted scoring module that maps the same features onto four health-risk dimensions and discuss the joint

output. The remainder of the paper is organised as follows: Section II reviews related work; Section III describes data collection, preprocessing, the learning pipeline, and the health-risk module; Section IV reports empirical results; and Section V concludes with limitations and planned extensions.

II. RELATED WORK

Research on problematic technology use splits along two broad lines: clinical and epidemiological work that quantifies prevalence and risk factors, and computational work that builds predictive tools on top of those findings. Our contribution sits in the second line but draws heavily on the first.

On the clinical side, Alsaleh [1] reviews how the diffusion of digital technology has restructured routine social behaviour, and Nilsson et al. [2] document a spike in gaming and social-media engagement among adolescents during the COVID-19 period that did not fully recede afterwards. A recent meta-analysis by Shiferaw et al. [3] pools effect sizes across youth studies and concludes that sustained overuse is associated with small-to-moderate reductions in sleep quality and academic achievement, together with elevated scores on depression and anxiety inventories. Qanash et al. [4] report similar patterns in a cross-sectional sample of health-care students, while Abi-Jaoude, Naylor, and Pignatiello [5] trace the mental-health implications of social-media saturation in younger cohorts. The screen-time–sleep link is further quantified by G et al. [6] and by Kolhar, Kazi, and Alameen [10], both of whom find that evening device use displaces sleep onset by a clinically relevant margin.

On the computational side, Lane et al. [7] investigated smartphone addiction alongside personality traits and sleep quality using standard statistical tools and noted the limits of rule-based scoring. Yarkoni and Westfall [8] make the broader methodological case for predictive modelling in psychology, arguing that classifiers optimised for out-of-sample accuracy can surface regularities that explanatory regression misses. Canoğulları [9] reviews psychotherapeutic interventions targeted at adolescent digital-behavioural disorders and highlights how little of that work is informed by individual-level prediction. Lozano-Blasco, Latorre-Martínez, and Cortés-Pascual [11] conduct a meta-analysis of internet addiction in adolescence and report substantial heterogeneity across instruments, which argues for flexible feature-based models. Leitner and Arendasy [12] compare classical machine-learning classifiers against neural networks on psychological tasks with modest sample sizes and find tree-based models competitive when the feature set is small and interpretable.

Across these references, two recurring gaps stand out. First, most predictive studies rely on a single algorithm or a single behavioural variable, so the relative contribution of different indicators is rarely reported. Second, addiction prediction and health-outcome estimation are almost always treated as separate pipelines, even though they draw on the same behavioural signals. The framework presented here addresses both points by benchmarking four classifiers on a shared feature set and by coupling the best classifier with a lightweight, interpretable health-risk scoring module that reuses those same features.

III. PROPOSED METHODOLOGY

A. Problem Setting

Digital addiction, in the operational sense used in this paper, refers to a pattern of smartphone and related-device use that the user cannot readily moderate and that begins to interfere with sleep, concentration, or academic commitments. Adolescents are singled out as the target population because their exposure is high, their coping resources are still developing, and the long-term effects of habits formed in this window tend to persist. We pose the screening problem as a three-way classification—low, moderate, or high risk—based on self-reported behavioural cues, and we pair the classifier with a continuous health-risk score that quantifies the likely physical and psychological cost of the observed usage pattern.

B. Behavioural Variables

Six behavioural features form the core input to the classifier. These are: (i) average daily screen time in hours; (ii) average daily social-media usage in hours; (iii) a binary indicator of phone use during the hour before sleep; (iv) the self-reported number of times the phone is unlocked per day; (v) average nightly sleep duration in hours; and (vi) a 1–5 ordinal rating of difficulty concentrating on academic tasks. These items were chosen because each has an independent literature base linking it to either addiction severity or a downstream outcome, and together they span the active (social media, checking), passive (screen time), and consequential (sleep, concentration) dimensions of device use.

C. Health-Risk Module

Alongside the addiction-risk classifier, we compute four health-risk indicators that are meant to be read by non-specialists. The four are eye-health risk, driven by total screen exposure and late-night device use; depression risk, driven by social-media exposure, restlessness items, and sleep loss; sleep-disorder risk, driven by night-time use and total sleep duration; and anxiety risk, driven by checking frequency, distraction ratings, and social-media exposure.

Each component is returned as a percentage using a simple weighted-sum formulation:

$$\text{Health Risk } \% = (w_1 \cdot F_1 + w_2 \cdot F_2 + w_3 \cdot F_3 + w_4 \cdot F_4) \times 100 \quad (1)$$

where $F_1 \ldots F_4$ are min–max normalised versions of the relevant behavioural features and the non-negative weights $w_1 \ldots w_4$ sum to one. Weights were fixed a priori from the clinical references reviewed in Section II rather than tuned on the dataset, so the score remains interpretable, and its rationale can be explained to a teacher or parent. This layer is deliberately transparent: the goal is explanation, not a second black-box prediction.

D. Data Collection

The dataset was assembled through a Google Form distributed to teenagers aged 13–19. A total of 125 complete responses were retained and exported to CSV. The instrument comprised three blocks: a block of behavioural items (daily screen time, social-media usage, pre-sleep phone use, checking frequency, sleep duration); a block of seven Likert-scale statements about compulsive use (for example, using the phone longer than intended, feeling restless without it, neglecting academic tasks, and being distracted during class), each rated on a five-point scale from 1 (least likely) to 5 (most likely); and a short academic block capturing current GPA and coarse demographic information. All processing was performed in Python 3.10 on Google Colab, using pandas for data handling, scikit-learn for modelling, and matplotlib together with seaborn for the visualisations reported in Section IV.

E. Preprocessing and Label Construction

The raw CSV was cleaned in four steps. Timestamp and other non-behavioural columns were dropped. Records with any missing value in the core feature set were discarded rather than imputed, to avoid injecting synthetic signal into a small sample. Categorical fields such as the pre-sleep phone-use indicator were label-encoded. The seven Likert-scale addiction items were summed into a composite Addiction Score, which was then binned into low, moderate, and high risk using tercile cut-points on the training fold so that class frequencies would be roughly balanced and comparable across folds.

F. System Workflow

The full pipeline runs in six stages: survey-based data collection; cleaning and encoding; Addiction-Score computation and risk-level labelling; classifier training; evaluation on a held-out test set; and finally, parallel computation of the health-risk scores for every respondent. Fig. 1 summarises the flow and highlights where the behavioural, academic, and health-risk signals feed into the final output.

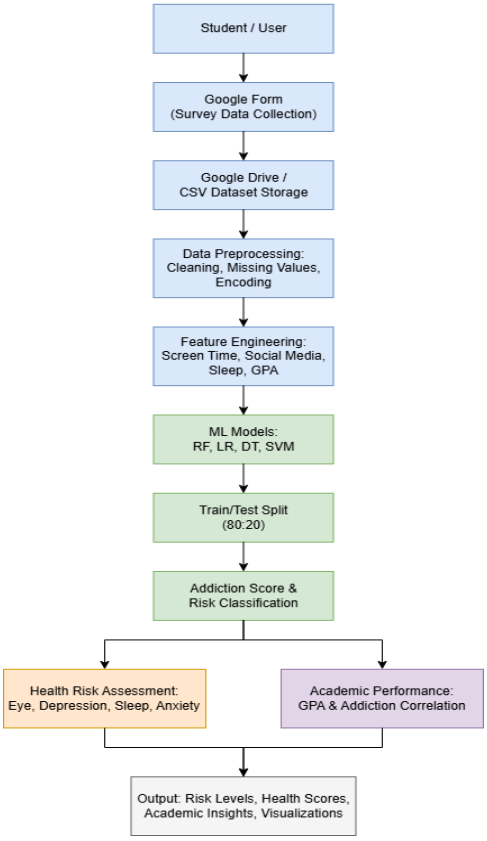

Fig. 1. End-to-end workflow of the proposed digital addiction and health-risk assessment framework.

The structure is intentionally modular. Replacing the classifier block with a different algorithm or retraining on a larger cohort does not change the upstream survey pipeline or the downstream health-risk overlay, which makes the system straightforward to extend.

G. Classification Models

Four classifiers were trained and compared: Random Forest (100 trees, Gini impurity, max depth unrestricted), Logistic Regression with L2 regularisation, Decision Tree (CART, pruned with minimum samples per leaf set to two), and Support Vector Machine with an RBF kernel and default class weighting. These four were chosen to span a reasonable spread of inductive biases—ensemble non-linear, linear, single-tree, and kernel—while keeping the models small enough that overfitting on a 125-row table is visible rather than hidden.

The dataset was split 80/20 into training and test partitions with a fixed random seed for reproducibility, stratified by the three-class label so that each band appeared in both partitions. The six behavioural features described in Section III-B were fed in without further feature engineering; the aim was to keep the feature space small and legible rather than to maximise accuracy through heavy transformations.

H. Evaluation Metrics

Performance is reported using accuracy, precision, recall, and F1 score, with per-class confusion matrices retained for error analysis. Accuracy follows the standard definition,

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (2)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives counted per class and then macro-averaged. Confusion matrices are used to diagnose where each classifier confuses adjacent risk bands, which is more informative in a three-class ordinal setting than headline accuracy alone.

IV. RESULTS AND DISCUSSION

The trained classifiers were evaluated on the 20% held-out test partition. Before turning to the comparative numbers, we first inspect the distribution of labels, since a skewed sample would colour every downstream metric.

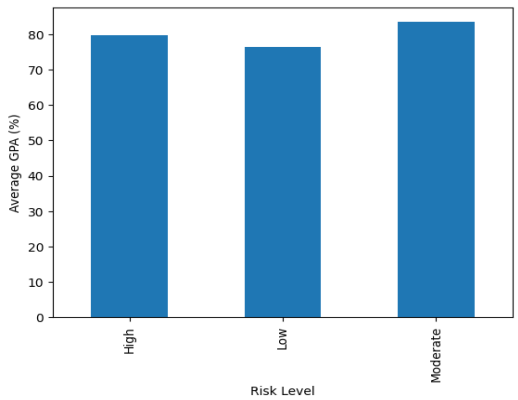

Fig. 2. Distribution of digital-addiction risk bands in the collected sample, shown alongside mean GPA per band.

As Fig. 2 shows, moderate-risk respondents make up the largest single group, with low-risk and high-risk bands flanking it in smaller but comparable numbers. This matches the pattern reported in the broader adolescent-screen-time literature, where the distribution of problematic-use scores is typically unimodal with a long right tail rather than bimodal. The mean GPA reading per band is noteworthy on its own: academic standing is not a monotonic function of risk band in this sample, and the highest-scoring students sit in the moderate band rather than the low band, which we return to in the discussion.

Table I lists headline metrics for all four classifiers.

TABLE I. Performance Comparison of the Four Classifiers.

Model	Accuracy	Precision	Recall	F1
Random Forest	0.75	0.76	0.74	0.75
Logistic Regression	0.625	0.63	0.62	0.62
Decision Tree	0.375	0.38	0.36	0.37
SVM (RBF)	0.625	0.64	0.61	0.62

Random Forest comes out on top on every metric, with 75% accuracy and a macro-averaged F1 of 0.75. Logistic Regression and SVM finish in a near tie at 62.5%, and the single Decision Tree trails well behind at 37.5%. The gap between the ensemble and the single tree is the clearest signal in the table: averaging over 100 bootstrapped trees roughly doubles the accuracy of a single CART learner on this sample, which is consistent with what one would expect when the feature–label relationship is mildly non-linear but the dataset is small. The bar chart in Fig. 5 makes the same comparison visually.

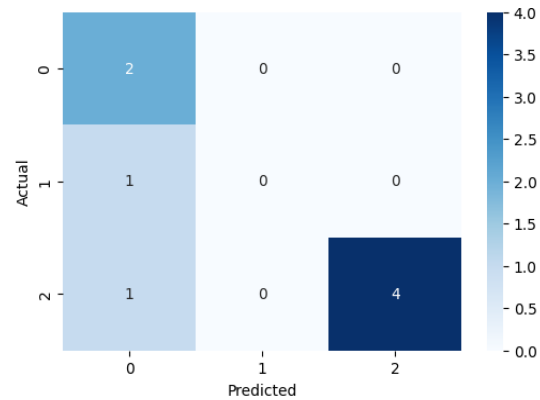

Fig. 3. Confusion matrix for the Random Forest classifier on the held-out test set. Rows are true labels, columns predicted labels; classes 0, 1, and 2 correspond to low, moderate, and high risk respectively.

The confusion matrix in Fig. 3 shows where the best model stumbles. It is confident on the extremes—it recovers the high-risk band (class 2) almost perfectly and correctly identifies the low-risk instances it sees—but it blurs the boundary between the moderate band and its neighbours. This is the failure mode we expected: moderate-risk teenagers share a large subset of behavioural cues with both of the adjacent classes, and with 125 respondents there simply is not enough data for the model to learn the finer decision surface reliably.

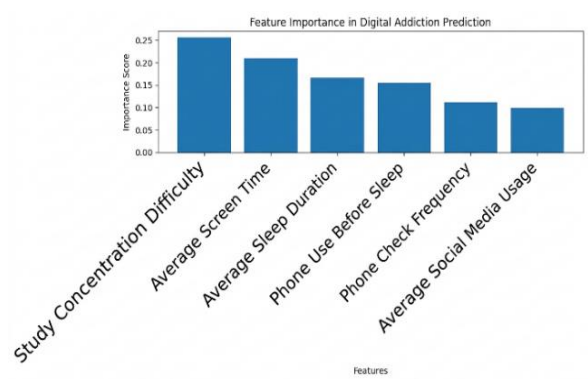

Fig. 4. Feature-importance scores from the Random Forest classifier. Higher values indicate a stronger contribution to the split decisions.

Feature importance from the trained forest (Fig. 4) is informative in its own right. Study-concentration difficulty tops the list, followed by average screen time, average sleep duration, and pre-sleep phone use; social-media usage and checking frequency contribute less than one might have

guessed from the descriptive literature. The ordering is worth dwelling on. Concentration difficulty is itself a downstream consequence of the other behaviours, and its high importance suggests the classifier is using it as a compact summary of several upstream variables rather than as an independent signal. In practical terms, a screening tool that asked only this item plus daily screen time would recover a large share of the discriminative power of the full questionnaire.

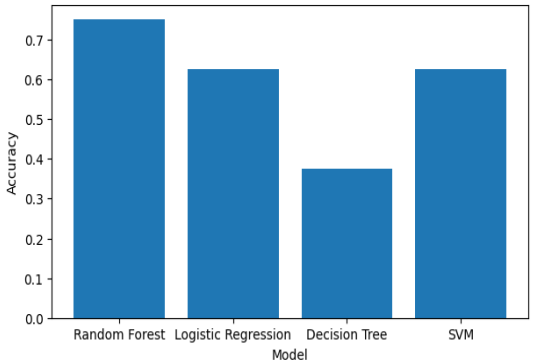

Fig. 5. Comparison of prediction accuracy across the four classifiers.

The model-comparison chart in Fig. 5 reiterates the numeric ranking. Random Forest is the clear leader; Logistic Regression and SVM are statistically indistinguishable given this sample size; and the single Decision Tree is substantially weaker, which is typical when a high-variance learner is trained on a small table with moderately correlated features.

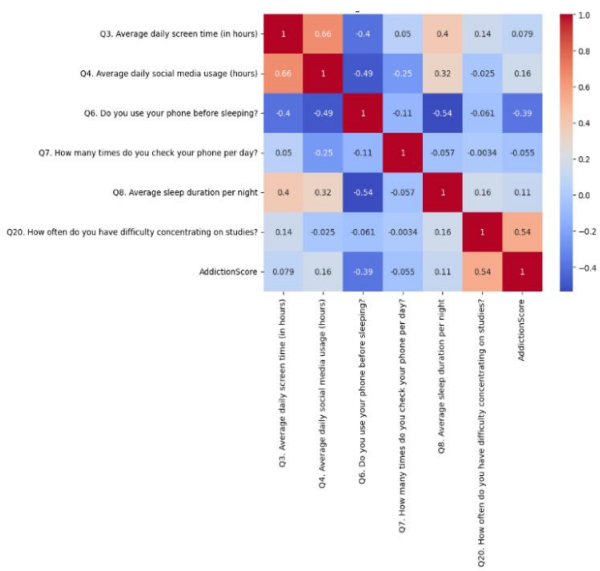

Fig. 6. Correlation heatmap of behavioural features and the composite Addiction Score.

The heatmap in Fig. 6 helps explain the feature-importance ordering. The Addiction Score correlates most strongly with concentration difficulty (+0.54) and, negatively, with pre-sleep phone use (−0.39 -noting that higher values of this field indicate the respondent avoids the phone before bed). Screen time and social-media exposure are tightly coupled (+0.66), confirming that they should not be treated as independent predictors, and sleep duration shows the expected negative correlation with night-time phone use (−0.54). These patterns are stable across repeated random splits.

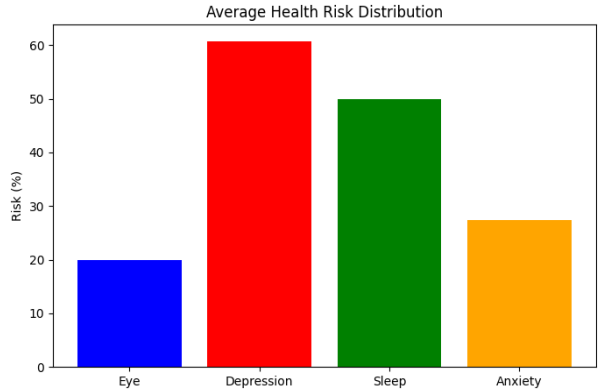

Fig. 7. Average health-risk distribution across the four indicators computed by the weighted-scoring module.

Turning to the health-risk overlay, Fig. 7 reports the mean of each of the four scores across the sample. Depression risk is the largest contributor at roughly 60%, sleep-disorder risk follows at about 50%, anxiety risk sits near 28%, and eye-health risk is lowest at around 20%. The pattern should be read with two caveats. First, these numbers reflect the weights fixed a priori from the clinical literature, not free parameters fitted to the data. Second, the eye-health figure is likely an under-estimate because screen-exposure variance in this cohort is narrow; a dataset with a wider range of total daily use would probably push this score up.

Finally, the relationship between the Addiction Score and self-reported GPA is weakly negative but noisy, and the earlier observation holds: the academically strongest respondents in our sample are in the moderate-risk band rather than the low-risk band. A plausible interpretation is that some level of device use is tightly bound up with contemporary study routines, i.e, online resources, group chats for homework, study-break content, and that the real academic cost kicks in only once usage crosses into the high-risk band. Confirming this would require a larger sample with a finer breakdown of how screen time is actually spent, which is on our list of follow-up work.

V. CONCLUSION AND FUTURE WORK

We set out to build a screening tool for adolescent digital addiction that is simple enough to deploy in a classroom setting, transparent enough to be read by a non-specialist, and more discriminating than the fixed-score questionnaires typically used in this space. The framework we report here does three things: it predicts addiction risk from a short behavioural survey using a Random Forest classifier, which reached 75% accuracy on the held-out test set; it ranks the behavioural cues by their contribution to that prediction, with concentration difficulty and daily screen time emerging as the dominant signals; and it overlays a weighted health-risk score across four domains (eye health, depression, sleep, anxiety), giving teachers and counsellors a compact summary of the

probable physical and psychological costs that accompany each risk band.

The limitations are worth stating plainly. A sample of 125 respondents from a single region is not large enough to generalise to the global adolescent population, and every measurement is self-reported, which introduces the usual biases. The health-risk weights are fixed rather than learned and should be validated against clinical outcomes in a future cohort. The classifier struggles on the boundary between the low and moderate bands, which is where early-intervention decisions are most sensitive.

Three extensions are planned. First, we will expand the questionnaire to a multi-institution cohort and release a larger anonymised dataset. Second, we intend to supplement self-reports with passively logged device-usage traces (daily unlocks, app-level screen time, bedtime cut-offs), which should reduce response bias and open the door to sequence models that use temporal patterns. Third, we will benchmark gradient-boosted ensembles and lightweight recurrent architectures against the tree-based baselines presented here, and evaluate whether explainability tools such as SHAP can make the predictions of those larger models as legible as the Random Forest feature-importance profile. Together, these steps should move the framework from a proof-of-concept prototype toward a screening instrument that is robust enough for real deployment in schools.

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